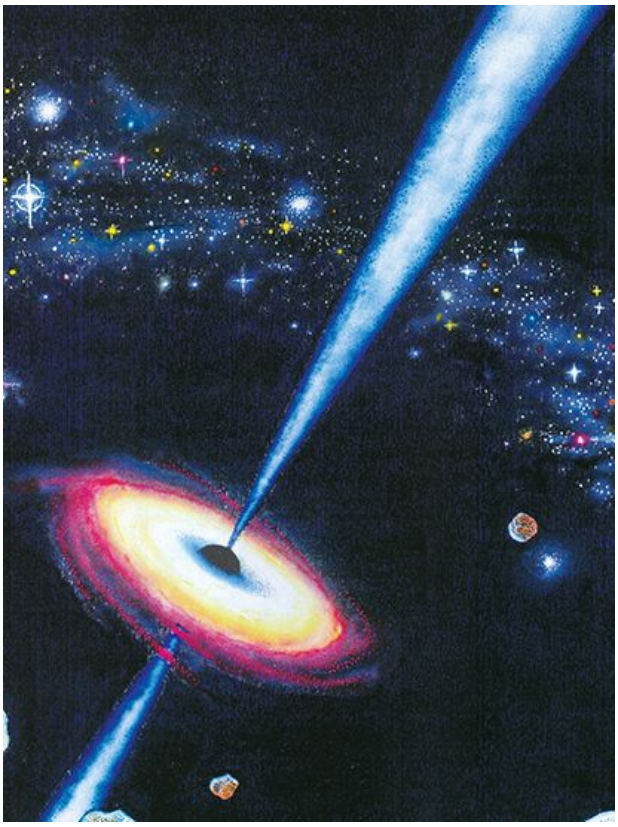


Fig. 9.2. Artist's conception of an accretion disk around a black hole, and jets emerging from near the hole's poles. *[Drawing by Matt Zimet based on a sketch by me; from my book* Black Holes & Time Warps: Einstein's Outrageous Legacy.*]*

Astrophysicists struggled for nearly a decade to explain how the jets are powered and what makes them so fast, so narrow, and so straight. The answers came in several variants, with the most interesting in 1977 from Roger Blandford at the University of Cambridge, England, and his student Roman Znajek, building on foundations laid by the Oxford physicist Roger Penrose; see Figure 9.3.

The accretion disk's gas gradually spirals into the black hole. When crossing the hole's event horizon, each bit of gas deposits its bit of magnetic field onto the horizon, and then the surrounding disk holds it there, Blandford and Znajek concluded. As the black hole spins, it drags space into whirling motion (Figures 5.4 and 5.5), and the whirling space